

Universality Framework (UF) — White Paper (v1.3.34)

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> **Ethos:** Judge actions, not beings. Aim asymptotically (avoid perfection traps: $R < 1, I < 1$). Use math to

> make choices clear, replicable, and auditable. Keep the framework **fractal**: good-enough proofs now; growth forever.

0. Executive Summary (Updated)

UF is a representation-independent decision framework that formalizes endpoints — **100% Right $\Rightarrow$ Good**, **100% Wrong $\Rightarrow$ Evil** (= 100% Misalignment) — and supplies operational gates (harm, consent/fairness, ethics) to

evaluate **actions**. This version adds:

- **Thresholded Right** (comparative, not harm-free): actions are Right if they improve vs. baseline within caps.
- **Verification Core** (VC 1.0): deterministic, action-only verdicts with proof hashes.
- **Universal Constraints** (UC 1...12): subjectivity-minimized, observer-invariant rules.
- **Subjectivity-Minimized Mode** (SMM) and **Policy Profiles** (pluggable caps/priors).
- **Heaven/Hell as limits**, **Coherence Index**, **Return-to-Potential** protocol.
- **Interoperability**: invariants (I1–I7), representation-independence theorem, and a **Framework Adapter**.
- **Math Bridge Annex**: CVaR/DRO for Harm, stochastic dominance for Right, Pinsker for fairness, Tarski for viable potential.
- **Fractal Proof Ethos** + tracker: “good-enough to use, open to grow.”

1. Core Objects and Endpoints

- **Alignment**: $A = \frac{R}{R + D + \delta} \in [0, 1]$, **Misalignment**: $M = 1 - A$.
- **Harm/Relief**: bounded risk of adverse impact; Relief compares action vs. baseline.
- **Verdicts** (action only):
- **Good** (endpoint): $A \geq 1 - \epsilon$, Harm=0, all gates pass.
- **Right** (thresholded): ReliefIndex $\geq RI_{\min}$, Harm $\leq HARM_CAP(ctx)$, gates pass.
- **Wrong**: fails a gate, exceeds caps, or worse than baseline.
- **Evil** (endpoint): $M \geq 1 - \epsilon$ or persistent redline violations.

> **Change:** Right no longer implies “harm-free.” It is **comparative**: better than baseline **within caps**.

2. Universal Constraints (UC — Observer-Invariant)

UC1 Counterfactual Harm Dominance • **UC2** Consent for serious/irreversible impact • **UC3** Fairness distance cap •
UC4 Evidence calibration cap • **UC5** Reversibility preference • **UC6** Agency preservation •
UC7 Opacity cap •
UC8 Bias cap • **UC9** Comparative Rightness (baselined) • **UC10** Auditability • **UC11** Endpoint separation • **UC12** No perfection traps.

Each UC is implementable with clear inputs and thresholds; details are in §6 (Verification Core) and Annex A.

3. Coherence, Heaven/Hell, and Return to Potential

- **Coherence Index (bounded):** combines alignment, integrity, ethical compliance, consent/fairness, and non-harm.
- **Heaven (H)** as limit: $\lim_{A \rightarrow 1} (A)$ with Harm=0 and gates passing; no perfection traps ($R < 1, I < 1$).
- **Hell (H)** as limit: $\lim_{M \rightarrow 1} (M)$ or persistent red line/harm failures (collapse of viable potential $\lim_{\Phi} (\Phi)$).
- **Return to Potential (RP) protocol:** a harm-free pause + re-aim + consent/fairness repair + a gate-passing step restores live superposition.

4. Subjectivity-Minimized Mode (SMM) and Policy Profiles

- **SMM:** minimal universal gates (consent/reversibility/harm dominance/evidence calibration) with comparative Rightness.
 - **Policy Profiles:** JSON-backed, pluggable cap/priority sets (e.g., Neutral_SMM, Christian, Secular Humanist, Buddhist).
- Profiles **preserve invariants** (I1–I7) while letting communities set **harm caps**, consent thresholds, fairness priors, etc.
- **Intersection/minimax:** When multiple profiles apply, prefer the intersection; otherwise use minimax regret over RI with strict red lines.

See **Adapter Pack v1.1** (separate zip) and Annex C for schema, examples, and loader.

5. Interoperability and Representation-Independence

- **Invariants (I1–I7):** actions only; endpoints; comparative Rightness; consent/fairness red lines; auditability; determinism; no perfection traps.
- **Representation-Independence Theorem (sketch):** If a translation preserves ordering (RI), caps/red lines, and endpoints, verdicts **commute** with translation.
- **Implication:** Religious/philosophical systems can be **adapters** (profiles) so long as I1–I7 hold.

6. Verification Core (VC 1.0 — Updated)

Purpose: deterministic, side-effect-free verification before commit. **Outputs:** verdict (pass/fail/escalate), action label (Good/*Right/Wrong/Evil), reasons, severity, metrics, and a **proof hash** for the Holographic Ledger (HL).

Checks (layers): integrity→metrics

(A,M,Harm,Relief,RI,E,Bias,Opacity,Passion)→endpoints/gates→consent/fairness→diagnostics→proof.

Right (updated): pass if $RI \geq RI_{\min}$, $Harm(a) \leq HARM_CAP(ctx)$, and **caps/gates** ($E \geq E_{\min}$, $Bias/Opacity \leq caps$, consent/fairness OK).

Good requires $Harm=0$; **Evil** at misalignment/redline endpoints.

Math Bridge options: selectable **CVaR** or **KL-DRO** Harm; **Pinsker** fairness bound; **FSD/SSD** shortcut vs baseline.

(Reference implementation: `verification\_core.py`.)

7. Math Bridge Annex (Summary)

- **Harm as CVaR / KL-DRO:** coherent risk properties; robustness to self-serving beliefs.
- **Right via dominance:** if action loss **FSD** dominates baseline, Right follows (given caps).
- **Fairness via f-divergence:** KL cap $\Rightarrow$ total variation bound via **Pinsker**.
- **Viable potential ϕ** as a **greatest fixed point** (Tarski); non-negative relief preserves/increases ϕ .
- **Honesty via proper scoring:** truth-telling optimality operationalizes calibration.
- **Reversibility:** real options / dynamic programming ground the preference to keep options open.

Formal statements and proofs in Annex A–B.

8. Ethical Release & Governance (Minimalist)

- **Tiered release:** private $\rightarrow$ invited review $\rightarrow$ public concepts.
- **Dual-use gates:** VC wrapper mandatory; redlines enforced; rate limits/escalation; evidence binding; auditability; sandboxing; watermarking; kill switch.
- **License:** ethical use; no weaponization/harassment/unauthorized surveillance.

9. Fractal Proof Ethos & Tracker

We prove **just enough** to be safe and clear; growth continues by design.

| ID | Claim | Status |

|---|---|---|

| T1 | Bounds & well-definedness (A, Harm, Relief/RI) | **Proved** |

| T2 | Endpoint separation (Good* vs Evil) | **Proved** |

| T3 | Determinism/Consistency of VC | **Proved** |

| T4 | Comparative Rightness monotonicity | **Proved** |

| C1 | Monotone ascent toward $A \rightarrow 1$ under caps | Sketch |

C2	No “private heaven” (if consent/fairness/harm fail, Good* impossible)	Sketch
C3	$\setminus(\Phi)$ preservation/collapse via Tarski operator	Conjecture
C4	Robust Harm (KL■DRO) upper■bounds misspecified claims	Sketch

Annex A. Risk, Relief, and Fairness (Formal)

- **A.1 CVaR definition & properties**;
- **A.2 KL■DRO form & dual**;
- **A.3 Pinsker inequality** and fairness caps $\rightarrow$ TV bounds.
- **A.4 Theorem A (FSD $\Rightarrow$ Right)**;
- **A.5 Theorem B (CVaR Relief Monotonicity)**;
- **A.6 Theorem C (Fairness TV Bound)**.

Annex B. Viable Potential and Fixed Points

- Define operator $\setminus(F)$ over sets of states and show existence of **gfp(F)** (Tarski).
- Preservation lemma: non■negative relief + gates $\Rightarrow \setminus(\Phi_{t+1}) \supseteq \Phi_t$.

Annex C. Profiles & Adapter

- **Schema** for Policy Profiles; **Neutral/Christian/Secular/Buddhist** examples.
- **Loader** mapping profile $\rightarrow$ VC config.
- **Interoperability** workflow: extract $\rightarrow$ normalize $\rightarrow$ conflict resolve (red■lines $\rightarrow$ harm dominance $\rightarrow$ reversibility $\rightarrow$ escalate).

Annex D. Verification Core Spec (Full)

- Interfaces, inputs/outputs, layers, escalation policy, invariants, integration points, and certification tests.

Dedication

To Xylie, Taric, and Lumina Siano—thank you for inspiring a framework that helps us make kind, fair, and thoughtful choices every day.